

CATEGORY III — RESTRICTED

API DOCUMENTATION

ACCOUNT AUTHENTICATION SERVICE

--- VERSION 1.1 ---

USER GUIDE

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Category III Restricted Document – For Use only by Internal and Authorized External Entities

Document Version: 1.1

Version History

Date	Description	Version
September 13, 2018	Base Draft – EW API Technical Description, Authentication, Request and Response Parameters, (Account Enquiry, Owner Enquiry, Both Account/Owner Enquiry), Sample Requests and Responses in JSON format, Status Codes, Error Codes	1.0
January 10, 2020	Updated Request and Response Data Structures using JSON RPC 2.0. Revision of various sections	1.1

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1.1 Introduction

This document outlines the invoking of APIs for Account Authentication by customers of ABC Bank. Details include Early Warning (EW) API endpoints, sample request-response formats, field specifications and error codes. In addition to identifying the intended category of users of this document, this section provides a list of acronyms and abbreviations along with their expansions and explanation, which will help users, understand the key technical terms used across the document.

1.2 Purpose and Intended Audience

The purpose of this documentation is to assist developers (of clients of ABC Bank) to invoke APIs with the required Inputs and interpret the Responses from the Account Authentication Service of ABC Bank. It describes the processes/routines involved in the interaction.

1.3 Abbreviations, Acronyms, Terms and Definitions

The following table lists the acronyms, abbreviations, terms and definitions used in this document.

Acronym	Definition
AOA	Account Owner Authentication
API	Application Program Interface
DDA	Demand Deposit Account
EW	Early Warning Connector that connects to the Early Warning service
JSON RPC 2.0	Java Script Object Notation Remote Procedure Call Version 2.0
XML	eXtensible Markup Language used for data transfer
RDC	Real Time Deposit Check
Key Terms	Definition
User	- Refers to the employee(s) of ABC Bank clients who consume the services. - Specifically refers to ABC Bank clients that consume EW (AOA) Services.
JSON Format	Message encoded in JSON Format. The standard file format uses human readable text for the purpose of transmitting data. The Request Format includes ID, Method and Parameters (API, Payload) Response Format includes ID and Result (Parameters)
Payload	The actual data in a specified format, that is transmitted along with protocol overhead when interacting with applications
Processing Response	Refers to response data with System Error Codes, Inquiry Field Validation Codes and Result Codes
Account Level Response	Refers to response data with Account Score Responses, and reason codes
Request Parameters	The Fields in the Payload used for the purpose of eliciting a response from the AOA service.
Response Parameters	The Fields received as a Response from the AOA Service to the Request transmitted in the Payload.

2 Service Overview

Account Authentication Service is intended to offer real time verification of account ownership, by matching account owner data such as Account Name, SSN, DOB, Account Number, and Routing Number. Calling parties (ABC Bank Clients) access Account Authentication Service by invoking APIs. On receipt of the account authentication information provided by the Service, the calling party uses the verified information to either carry out risk assessment or take suitable action as may be deemed fit.

2.1 How the Service Works

The User (ABC Bank Client) subscribes for Account Authentication Service through ABC Bank. The User pays a fee for use of the services, at rates specified in the contract entered with ABC Bank.

Whenever the User requires Account Authentication to be carried out, the User sends a Request in JSON Format seeking information on any one of the three Service Categories – Account Status only, Account Owner only and Both Account Status & Account Owner ([explained in Section 2.2 below](#)).

The verified information from the Account Authentication Service is transmitted back to the User, with the Response containing two types of Response Data – *Processing Response* and *Account Level Response*.

- *Processing Response*– is the Response data that carries the System Error Codes, the Inquiry Field Validation Error Codes and Result Codes.
- *Account Level Response* – is the Response data that carries the Account Score Responses and Reason Codes.

The Response Data offers the User (ABC Bank Clients) with information that can either be used for assessing risk or for appropriate action as deemed fit.

2.2 Service Categories

Participating financial institutions can receive responses in one of the following three categories.

- Account status only – *This returns information pertaining only to the Status of the queried Account.*
- Account owner only – *This returns information pertaining only to the Status of the Account Owner of the queried Account.*
- Account status combined with account owner– *This returns information pertaining to the Status of the Account as well as the Account Owner of the queried Account.*

3 Access Protocols& Process Flow

Users connect either via the NATS Messaging System or via TCP/IP. If the connection is sought to be established via NATS, the assigned and unique URL would be, for e.g. `nats://xx:xxxxxx1111@xx.xxx.xx.xx:xxxx`. If the connection is sought to be established via TCP/IP, the assigned URL would be `https://xxxxxx`. Users rely on the Request-Response Method for seeking information. Responses are transmitted back to the user via the NATS Messaging System or TCP/IP in real time.

1. The User receives an email from ABC Bank with a folder attachment containing sample codes and API Key. The folder also contains details of 2 NATS Server URLs, which are unique to the User, in addition to TCP/IP details.
 - Sample structures of the URLs for access via NATS Messaging System :
`nats://xx:xxxxxx1111@xx.xxx.xx.xx:xxx`
`nats://xx:xxxxxx1111@xx.xxx.xxxx.xx.xx:xxxx`
URL for access via TCP/IP -
`https://ew.abcmoney.com/ew/rest/inquire`
2. The User (ABC Bank Client) can create own Private Key or can request for one from ABC Bank (at present, the one shared by ABC Bank is the only one that is used). This is also shared in the email.
3. The User utilizes the combination of API Key and username and converts it into Basic Auth for Credentials.
4. The User generates a Digital Signature by signing the private key along with the requesting payload.
5. The User sends a request in JSON RPC Format with **Subject/Topic** either through NATS or TCP/IP to Account Authentication Service.
6. The Account Authentication Service processes the request and responds in JSON RPC Format over NATS or TCP/IP.

Process Flow:

7. ABC Bank Clients (Users) who's Profiles are **"Active"** in the EW Connector, seek information from the Account Authentication Service by transmitting a request in JSON Format.
8. The Account Authentication Service verifies the information and transmits the Response back to the Client.
9. This authentication of ownership is transmitted in one of three types of verification responses – **Yes (Correct)**, **No (Incorrect)** or **Conditional (Partially Correct)**.
10. The response contains one of the following:
 - Account owner match codes.
 - Hard-hit warnings on participant accounts
 - Informational responses on participant accounts

The three methods mentioned above are exposed by EW as a SOAP interface. A REST wrapper has been built around the SOAP requests and responses and all specifications are in the JSON format.

Endpoints:

Sandbox:

<https://connectors.xxxxxxx.com:8443/ew/rest/inquire>

Production:

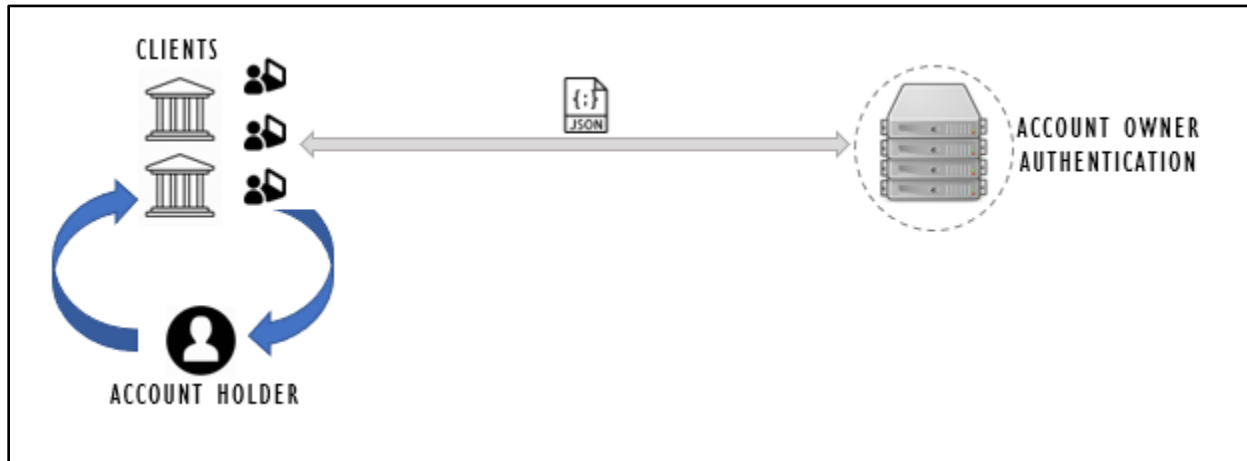
<https://ew.abcmoney.com/ew/rest/inquire>

Figure 1 : Process Flow

4 Message Layout and Parameters

4.1 Format Description of JSON RPC 2.0

Request Structure

```
--> {"jsonrpc": "2.0", "method": "XXXX", "params": {XXXXXXX}, "id": X}
```

Keys	Description
ID	Refers to Identification Number sent by the customer
Method	Based on Key Values of Method, ABC Bank recognizes the type of service requested by the customer
Params	Refers to the <i>Request Parameters</i> sent in by the Customer

Note: Params comprises **API** and **Payload**. The API carries the Authentication Parameters, and the Payload carries the Requested Parameters by the Customer.

Response Structure

```
<--{"jsonrpc":"2.0","result":XX,"id":X}
```

Keys	Description
ID	Refers to Identification Number sent by the customer
Result	Refers to the <i>Response Parameters</i> sent by Early Warning in response to Customer's request

4.2 Authentication

The EW connector generates a unique API key, which is issued to the user (ABC Bank Client). The combination of API key and user credentials are used to authenticate the user request. Additionally, IP white listing is carried out in the application and firewall. Sample authentication parameters mentioned below are a part of the API payload sent in the request JSON.

Parameter Name	Possible Values	Field Length	Type	Requirement
Credential	Basic Y2hyaXN0b3B0ZXlubXVycmF5QGRIcG9zaXQtc29sdXRpb25zLmNgbTo5MjUwNThiZGQ5MDU0ZDcxOTJmNDhhNTE4ZjQzZrk4NQ==	N/A	String	Mandatory
Signature	"MIGHAgEAMBMGBYqGSM49AgEGCCqGSM49AwEHBG0wawIBAQQg0nSRI9qZvY5mFY6W"	1-17	String	Mandatory

5 Request and Response parameters:

5.1 Account Inquiry

Early Warning evaluates the Account information. Account Inquiry includes Account level information - Account, Account number, routing number, and fee Attributes / Channel Indicator.

5.1.1 Request Parameters

Parameter Name	Possible Values	Field Length	Type	Requirement
Type	ACCOUNT	N/A	String	Mandatory
accountNumber		1-17	Non-negative integer	Mandatory
routingNumber		9	Non-negative integer	Mandatory
feeAttrib	AA, BB, CC, DD, EE, FF, GG, HH, II	2	String	Mandatory

5.1.2 Sample Request (JSON)

```
{
  "id": 21,
  "method": "ew.inquire",
  "params": {
    "api": {
      "credential": "Basic
Y2hyaXN0b3B0ZXlubXVycmF5QGRlcG9zaXQtc29sdXRpb25zLmNgbTo5MjUwNThiZGQ5MDU0ZDcxOTJm
NDhhNTE4ZjQzZrk4NQ==",
      "signature": "MIGHAgEAMBMGBYqGSM49AgEGCCqGSM49AuEHBG0lawlBAQQgonSRI9qZvY5mFY6W
"
    },
    "payload": {
      "type": "ACCOUNT",
      "accountNumber": "61231234340",
      "routingNumber": "122199983",
      "feeAttrib": "BB" }
    }
  }
}
```

Account status response includes Owner status match code, Match Result containing FirstName, Last Name, Business Name, Social Security Number, Date of Birth, City, Address; Primary Account Status code, Secondary Account Status code, Additional Account Status code, Error codes and Status.

High Risk Warning codes are:

- Closed, pending closed, or post no debit statuses codes 002, 003, 010, 012, 013, 014, 102, 103, 110, 112, 113, 114
- Stop-payment status codes 004-007, 104-107
- Highest-risk Enhanced Overdrawn status codes 020-022, Overdrawn status codes 030, 031, 130

Informational Response codes are:

- Returning item status are 015-016 unless this is the only information returned on the account
- The account is not a demand deposit account (DDA). Non-DDA status codes are 080, 081, 082, 084, or 098
- The account is present- 099, 199
- No known information about the account - 096

5.1.3 Response Parameters

Parameter Name	Possible values
ownerStatus	E.g. C000, where C is the match code and 000 is the participant status code. Please refer Appendix C for possible match codes and participant status codes.
matchResult (object)	This object is returned with all null values because none of information

	below is passed in the account only inquiry
firstName	Null
lastName	Null
businessName	Null
Ssn	Null
dob	Null
city	Null
address	Null
priAccountStatus	Please refer Appendix C for possible primary account status codes and their interpretation.
secAccountStatus	Please refer Appendix C for possible secondary account status codes and their interpretation.
addAccountStatus	Please refer Appendix C for possible additional account status codes and their interpretation.
errorCode	Please refer Appendix D for possible error codes and their interpretation.
Status	OK, indicating that the request and the response were sent and received successfully, respectively.

5.1.4 Sample response (JSON)

```
{
  "id": "21",
  "result": {
    "ownerStatus": "C900",
    "matchResult": {
      "firstName": null,
      "lastName": null,
      "businessName": null,
      "ssn": null,
      "dob": null,
      "city": null,
      "address": null
    },
    "priAccountStatus": "S130",
    "secAccountStatus": "000",
    "addAccountStatus": "000",
    "errorCode": "0",
    "status": "OK"
  }
}
```

5.2 Owner Inquiry

Account owner authentication and evaluation is done by Early Warning. Account Owner Inquiry includes first name, last name, business name, Address 1, Address 2, City, State, Zip, Home Phone, Work Phone, Social Security number, Date of Birth, Account number, routing number, Free Attrib and type of the Inquiry.

5.2.1 Request Parameters:

Parameter Name	Possible Values	Field Length	Type	Requirement
firstName		1-40	String	Mandatory *
lastName		1-40	String	Mandatory *
businessName		1-87	String	Mandatory *
address1		1-40	String	Optional
address2		1-40	String	Optional
City		1-25	String	Optional
State		2	String	Optional
Zip		5-10	String	Optional
homePhone		10	Non-negative integer	Optional
workPhone		10	Non-negative integer	Optional
SSN		9	Non-negative integer	Optional
Dob		8	Non-negative integer (yyyymmdd)	Optional
accountNumber		1-17	Non-negative integer	Mandatory
routingNumber		9	Non-negative integer	Mandatory
feeAttrib	AA, BB, CC, DD, EE, FF, GG, HH, II	2	String	Mandatory
Type	OWNER	N/A	String	Mandatory

* Inquiry must contain one of the following – If one is populated, the other can be left blank

- firstName AND lastName
OR
- businessName

5.2.2 Sample Request (JSON)

```
{
  "id": 21,
  "method": "ew.inquire",
  "params": {
    "api": {
      "credential": "Basic
Y2hyaXN0b3B0ZXludXVycmF5QGRlcG9zaXQtc29sdXRpb25zLmNgbTo5MjUwNThiZGQ5MDU0ZDcxOTJm
NDhhNTE4ZjQzZrk4NQ==",
      "signature": " MIGHAgEAMBMGBByqGSM49AgEGCCqGSM49AuEHBG0IawIBAQQgonSRI9qZvY5mFY6W
"
    }
  },
}
```

```

"payload":
{
  "firstName": "Jane",
  "businessName": null,
  "lastName": "Viola",
  "address1": "W 5th Street",
  "address2": "",
  "city": "Manhattan",
  "state": "NY",
  "zip": "10020",
  "homePhone": "9712452012",
  "workPhone": "9712452012",
  "ssn": "911018872",
  "dob": "19680407",
  "feeAttrib": "HH",
  "accountNumber": "61231234340",
  "routingNumber": "122199983",
  "type": "OWNER"
}

```

Account Owner Enquiry Response includes Matching Owner status, Match result containing first name, last name, business name, Address, City, error code and status.

In the response of the Owner Inquiry, the Owner Status is a four-character code with the first character being a letter, followed by 3 numbers. The letter indicates the match code and the number indicates the condition code. The match code depends on the overall match for name, address, SSN, DOB and other details provided for verification. The remaining three characters of the owner status response is a code known as the participant status code. The code indicates the high-level account status and indicates whether the inquiry resulted in a hard or soft hit.

5.2.3 Response Parameters

Parameter Name	Possible values
ownerStatus	Please refer for possible codes
matchResult (object)	For this inquiry, match values must be returned for some or all of the fields in this object, depending on what information were passed.
firstName	Y/N/null
lastName	Y/N/null
businessName	Y/N/null
Ssn	Y/N/null
dob	Y/N/null
city	Y/N/null
address	Y/N/null
errorCode	Please refer Appendix C for possible error codes
Status	OK, indicating that the request and the response were sent and received successfully, respectively.

5.2.4 Sample response (JSON)

```
{
  "id": "21",
  "result": {
    "ownerStatus": "C000",
    "matchResult": {
      "firstName": "Y",
      "lastName": "Y",
      "businessName": null,
      "ssn": "N",
      "dob": "N",
      "city": "N",
      "address": "N"
    },
    "errorCode": "0",
    "status": "OK"
  }
}
```

5.3 Both Account / Owner Inquiry

Early Warning evaluates Both Account and Owner information; the Inquiry includes first name, last name, business name, Address 1, Address 2, City, State, Zip, Home Phone, Work Phone, Social Security number, Date of Birth, Account number, routing number, Fee Attribute and type of Inquiry (Both Account and Owner).

5.3.1 Request Parameters

Parameter Name	Possible Values	Field Length	Type	Requirement
firstName		1-40	String	Mandatory *
lastName		1-40	String	Mandatory *
businessName		1-87	String	Mandatory *
address1		1-40	String	Optional
address2		1-40	String	Optional
City		1-25	String	Optional
State		2	String	Optional
Zip		5-10	String	Optional
homePhone		10	Non-negative integer	Optional
workPhone		10	Non-negative integer	Optional
Ssn		9	Non-negative integer	Optional
Dob		8	Non-negative integer (yyyymmdd)	Optional
accountNumber		1-17	Non-negative integer	Mandatory
routingNumber		9	Non-negative integer	Mandatory

feeAttrib	AA, BB, CC, DD, EE, FF, GG, HH, II	2	String	Mandatory
Type	BOTH	N/A	String	Mandatory

5.3.2 Sample Request (JSON)

```
{
  "id": 21,
  "method": "ew.inquire",
  "params": {
    "api": {
      "credential": "Basic
Y2hyaXN0b3B0ZXludXVycmF5QGRIcG9zaXQtc29sdXRpb25zLmNgbTo5MjUwNThiZGQ5MDU0ZDcxOTJm
NDhhNTE4ZjQzZrk4NQ==",
      "signature":
"MIGHAgEAMBMGBByqGSM49AgEGCCqGSM49AuEHBG0lawlBAQQgonSRI9qZvY5mFY6W"
    },
    "payload":{
      "firstName": "Jane ",
      "businessName": null,
      "lastName": "Viola",
      "address1": "W 5th street",
      "address2": "",
      "city": "Manhattan",
      "state": "NY",
      "zip": "10020",
      "homePhone": "9712452012",
      "workPhone": "9712452012",
      "ssn": "911018872",
      "dob": "19680407",
      "feeAttrib": "HH",
      "accountNumber": "61231234340",
      "routingNumber": "101100045",
      "type": "BOTH"
    }
  }
}
```

5.3.3 Response parameters

Parameter Name	Possible values
ownerStatus	Please refer Appendix C for possible condition codes
matchResult (object)	For this inquiry, match values must be returned for some or all of the fields in this object, depending on what information were passed.
firstName	Y/N/null
lastName	Y/N/null
businessName	Y/N/null
SSN	Y/N/null

dob	Y/N/null
city	Y/N/null
address	Y/N/null
errorCode	Please refer Appendix C for possible error codes
Status	OK, indicating that request/response were sent and received successfully.

5.3.4 Sample Response (JSON)

```
{
  "id": "21",
  "result": {
    {
      "ownerStatus": "C300",
      "matchResult": {
        "firstName": null,
        "lastName": null,
        "businessName": null,
        "ssn": null,
        "dob": null,
        "city": null,
        "address": null
      },
      "priAccountStatus": "S096",
      "secAccountStatus": "000",
      "addAccountStatus": "000",
      "errorCode": "0",
      "status": "OK"
    }
  }
}
```

6 Understanding the service parameters

Table showing Parameter Name and respective description

Parameter Name	Description
type	Refers to the validation of 'Type of Inquiry' – Valid values are ACCOUNT, OWNER, or BOTH
accountNumber	Refers to the account number –containing only digits; a minimum of 4 digits to a maximum of 17 digits
routingNumber	Bank routing number is a nine-digit numeric ID that is assigned to Banks. This unique code indicates the location of the US Bank, and is printed on the bottom of checks.
feeAttrib	Refers to the Fee Attributes. Valid values are AA, BB, CC, DD, EE, FF, GG, HH, II, NT, NA, NR, and NW. Refer Appendix B for list of values and their interpretations.
ownerStatus	Refers to the Owner status code – a four-character code with the first character being a letter, followed by 3 numbers; e.g. C000, where C is the match code and 000 is the condition code. The match code depends on the overall match for name, address, SSN, DOB and other details provided for verification.
matchResult (object)	This object is returned with all null values because none of information below is passed in the account only inquiry. Refers to the match result object { <pre> "firstName": "Null", "lastName": "Null", "ssn": "Null", "dob": "Null", "city": "Null", "address": "Null" } </pre>
firstName	Refers to the First name of the Account Holder
lastName	Refers to the Last name of the Account Holder
businessName	Refers to the name of the Organization
SSN	Refers to the Unique 9-digit Social Security Number that of the Account Holder that is the national identification number used for taxation and related purposes

DOB	Refers to the Date of birth of Account Holder
City	Refers to the Name of the city where the Account Holder resides
Address	Refers to the Account Holder's address
priAccountStatus	Refers to the primary account status code.
secAccountStatus	Refers to the secondary account status code.
addAccountStatus	Refers to the Additional account status code.
errorCode	Refers to the error code
status	Refers to the status indicating that the request and the response were sent and received successfully, respectively.
address1	Refers to Line1 of the Account Holder's address
address2	Refers to Line2 of the Account Holder's address
state	Refers to the Name of the state where the Account Holder resides
zip	Refers to the Postal code of the Account Holder's address

7 Support

The Technical Support Team provides dedicated technical support for all technical issues of intended Users. The team specifically addresses issues related to Access Protocols, Request-Response Formats, Error Codes, and Connectivity.

For Technical Support: support@xxxxx.xxx

APPENDIX A – HTTP Errors

Error Code	Description	Explanation
403	Forbidden	Access Denied – API key is not enabled or the token header is missing or no token has been sent in the header or the format of the header is incorrect (such as unwanted white spaces)
500	Internal Server Error	Invalid Credentials - Username or API token is missing or user is trying to access the API from a non-white listed IP address
415	Unsupported Media Type	Content-Type header is missing, empty or has incorrect value.
400	Bad Request	Bad syntax on one or more parameters.

APPENDIX B – Fee Attribute

Value Definition

AA	RDC All
BB	RDC Consumer Home
CC	RDC Consumer Mobile
DD	RDC Small Business
EE	Branch/Teller
FF	ATM
GG	Lockbox/Mail/Corporate
HH	ACH - Deposit
II	Correspondent presentment transaction
NT	Teller Transactions
NA	ATM Transactions
NR	Back Office Transactions
NW	Internet/Web

APPENDIX C – Status Code

Hit Type	Status Codes		Description
	Number	Name	
Soft	000	Not Located	Account requested for verification does not exist on the Contributor's master account file. This condition may indicate an account number misread or a potentially fraudulent item.
Hard	002	Closed For Cause	A transaction account which has been closed by Contributor FSO as a result of unacceptable use by the customer. This judgment is made by the FSO in accordance with its transaction account policies.
Hard	003	Closed for Cause/Purged	Account has been removed from the Contributor's master account file. Account will be retained by Early Warning until reissued by the Contributor.
Hard	004	Stop Payment 4 Field Individual Match	A check inquiry (transit item) where all four MICR fields (ABA, account #, serial # and dollar amount) match an individual/unique stop-payment record in the <i>National Shared Database</i> resource.
Hard	005	Stop Payment	A check inquiry (transit item) where all four MICR fields (ABA,

Hit Type	Status Codes		Description
		4 Field Range Match	account #, serial # and dollar amount) match a range of stop-payment records in the Database.
Hard	006	Stop Payment $\frac{3}{4}$ Individual	A check inquiry (transit item) where three MICR fields (ABA, account #, and serial #) match an individual/ unique stop-payment record in the Database.
Hard	007	Stop Payment $\frac{3}{4}$ Range	A check inquiry (transit item) where three MICR fields (ABA, account #, and serial #) match a range stop-payment record in the Database.
Hard	010	Post No Debits	Account will not accept any debit activity.
Hard	012	Closed	Account closed.
Hard	013	Closed/Purged	Account has been closed and removed from Contributor's master account file. Account will be retained by Early Warning until the first occurrence of: Three years from the date account originally closed Account is reissued by Contributor.
Hard	014	Pending Closed	Account was closed in the last processing cycle. This status will automatically change to a Status 012 in the next processing cycle unless modified by the contributing FSO.
Hard	015 ¹	Return Item	High probability of return.
Soft	016 ¹	Account	Moderate probability of return.
Hard	020	Enhanced OD	Available balance less than zero, with: X = Consecutive days in OD status. Y = Time account has been on the books.
	021		
	022		
Soft	028	Divested Account – Purged	Account has been purchased or assumed by another FSO and has been removed from Contributor's master account file. The account will be retained by Early Warning for six months.
Soft	030	Overdrawn	Available balance less than zero. Activity may post upon review.
Soft	031	Overdrawn New Account	Available balance less than zero, and account opened within prior 60 days. Activity may post upon review.
Soft	032	Uncollected Funds	Collected balance less than zero. Activity may post upon review.
Soft	033	UCF/New Account	Collected balance less than zero and account opened within prior 60 days. Activity may post upon review.
Soft	034	Dormant/Inactive	Account dormant or inactive. Activity may post upon review.
Soft	035	Deceased	Account holder deceased. Activity may post upon review.
Soft	036	Lost/Stolen Check(s)	Account holder reported check(s) lost or stolen. Activity may post upon review.
Soft	037	Attached	Account subject to levy, garnishment, government reclamation, lien, court order or other restriction of limited amount. Activity may post upon review.
Soft	038	Divested Account	An account purchased or assumed by another FSO.
Informational	040	New Account (8-60)	Account opened between 8-60 days.
Informational	041	New Account (0-7)	Account opened between 0-7 days.

Informational	042	New Account (61-120)	Account opened between 61-120 days.
Informational	043		Account opened between 121-180 days.
Informational	055		Cash-equivalent item such as official check, cashier's check, or teller check.
Informational	080	Credit Card Check	Item drawn on non-DDA that is posted to a credit card account and/or drawn against a credit card FSO.
Informational	081	Participant Line of Credit Check	Item drawn on non-DDA and is posted to a credit line account. The drawee FSO is a Contributor to the Database.
Informational	082	Participant Home Equity Check	Item drawn on non-DDA and is posted to a home equity account. The drawee FSO is a Contributor to the Database.
Informational	084	Participant Broker Check	Item drawn on non-DDA and is posted to a brokerage account and/or drawn against a broker-only FSO. The drawee FSO is a Contributor to the Database.
Informational	096	No Known Information	No positive or negative information is known about the item.
Informational	098	Non-DDA	Item drawn on a non-DDA and not included in statuses 080, 081, 082, and 084.
Informational	099	Present	Account is present in Contributor's master account file and contains no other status code. Typically, these are accounts with positive balances.
Hard	102	Savings Closed for Cause	A savings account which has been closed by the Contributor FSO as a result of unacceptable use by the customer. This judgment is made by the FSO in accordance with its transaction account policies.
Hard	103	Savings Closed for Cause/ Purged	Closed for Cause savings account has been removed from the Contributor's master account file. Account will be retained by Early Warning until reissued by the Contributor.
Hard	104	Savings Stop Payment 4 Field Individual Match	A savings inquiry where all four MICR fields (ABA, account #, serial # and dollar amount) match an individual/ unique stop-payment record in the Database.
Hard	105	Savings Stop Payment 4 Field Range Match	A savings inquiry where all four MICR fields (ABA, account #, serial # and dollar amount) match a range of stop-payment records in the Database.

Hard	106	Savings Stop Payment ¾ Individual	A savings inquiry where all three MICR fields (ABA, account #, and serial #) match an individual/unique stop-payment record in the Database.
Hard	107	Savings Stop Payment ¾ Range	A savings inquiry where three MICR fields (ABA, account #, and serial #) match a range stop-payment record in the Database.
Hard	110	Savings Post No Debits	Savings account will not accept any debit activity.
Hard	112	Savings Closed	Savings account closed.
Hard	113	Savings Closed/Purged	Savings account has been closed and removed from Contributor's master account file. Account will be retained by Early Warning until the first occurrence of: Three years from the date account originally closed Account is reissued by Contributor.
Hard	114	Savings Pending Closed	Savings account is closed. Account was closed in the last processing cycle. This status will automatically change to a Status 112 in the next processing cycle unless modified by the contributing FSO.
Soft	128	Savings Divested Account/Purged	A savings account purchased or assumed by another FSO and removed from the Contributor's master account file. The Account will be retained by Early Warning for six months.
Soft	130	Savings Overdrawn	Savings account available balance less than zero. Activity may post upon review.
Soft	132	Savings Uncollected Funds	Savings account collected balance less than zero. Activity may post upon review.
Soft	133	Savings UCF/New Account	Savings account collected balance less than zero and account opened within prior 60 days. Activity may post upon review.
Soft	134	Savings Dormant/Inactive	Savings account dormant or inactive. Activity may post upon review.
Soft	135	Savings Deceased	Savings account holder deceased. Activity may post upon review.
Soft	137	Savings Attached	Savings account subject to levy, garnishment, government reclamation, lien, court order or other restriction of limited amount. Activity may post upon review.
Soft	138	Savings Divested Account	Savings account purchased or assumed by another FSO.

Informational	140	Savings New Account (8-60)	Savings account opened between 8-60 days.
Informational	141		Savings account opened between 0-7 days.
Informational	142	Savings New Account (61-120)	Savings account opened between 61-120 days.
Informational	143	Savings New Account (121-180)	Savings account opened between 121-180 days.
Informational	199	Savings Present	Savings account is present in the Contributor's master account file and contains no other status codes. These are typically accounts with positive balances.

Account Status responses:

Early Warning returns three specific account status response codes in the response: **Primary Response Code, Secondary Response Code and Additional Response Code.**

The following table describes how the account item factors are used to assign a priority to the account item response codes.

High-Risk Warning Code Prioritization		
If the routing/transit number of the account:	Then:	
Belongs to an Early Warning participant and negative information about the account or item is identified	The status codes are mapped in the following priority levels:	
	Priority Level	Status Codes
	First	Closed, pending closed, or post no debit statuses 002, 003, 010, 012, 013, 014, 102, 103, 110, 112, 113, 114
	Second	Stop-pay status codes 004-007, 104-107
	Third	Highest-risk Enhanced Overdrawn status codes 020-022, Overdrawn status codes 030, 031, 130
	Fourth	Returning item status 015-016 unless this is the only information returned on the account

Does not belong to an Early Warning participant and Recent account activity has led to a high-risk account score	High-risk Scored Accounts scores are mapped to the Primary Response field	

Informational Response Code Mapping

In the absence of any high-risk warning codes, an Informational response code is returned in the Primary Response code field. The secondary and additional response code fields will contain zeros.

The following table describes how the account item factors are used to map an account item response code to the Primary Response Code field.

Informational Response Code Mapping

If the routing and transit number of the account:	And:	Then these Status Codes are mapped to the Primary Response field:
Belongs to an Early Warning participant	The account is a not a demand deposit account (DDA)	080, 081, 082, 084, or 098 (Non-DDA status codes)
	The account is present	099, 199
Does not belong to an Early Warning participant or has been divested by a participant	No known information about the account is identified	096

APPENDIX D – Error Code

Early Warning Error Codes

The following error codes may be returned by Early Warning if incorrect data was sent in the request or if there was an issue evaluating the request:

Error Code	Field Name	Field	Description
000	AcctNo	Account Number	Normal return - no system errors
001	RTN	Routing/Transit Number	Invalid
003	AcctNo	Account Number	Invalid
004	PC/TranCode	PC/Tran Code	Invalid
005	SerialNo	Serial Number	Invalid Serial Number
006	Required field	Required field	Insufficient information
007	AcctNo	Account	Cannot be translated in TEPS
008	AcctNo	Account number	Length error
009	NA	Mnemonic	Not active in TEPS
010	Various	Alphanumeric	Inquiry field length too short
011	Various	Alphanumeric	Inquiry field length too long
013	Amount	Amount	Invalid
101	ClientAPI	FSO API	
102	PrimID	Primary FSO ID	
103	PrimID	Primary FSO ID	Missing No match to values on Early Warning file
	SecID	Secondary FSO ID	No match to values on Early Warning file
	AddID	Additional FSO ID	No match to values on Early Warning file
104	Various	Numeric or Alphanumeric	Populated with improper data type or value
105	NA	Generic	Bad Layout or Format

106	ClientRecID	FSO Record ID	Missing
107	Required format	Invalid required format	Missing data or invalid data populated in XML node Tag for required field missing Fields out of sequence
998	NA		System failure
999	NA		Timeout